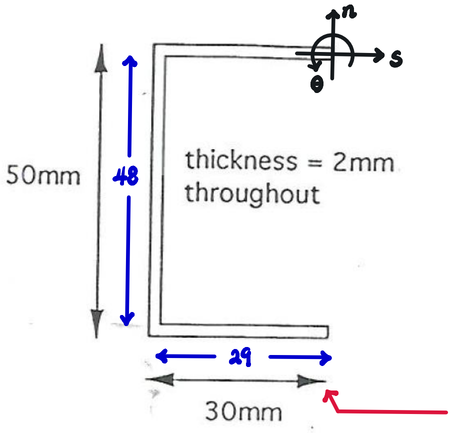
Structures 3

Tutorial Sheet 1

1.

a) The channel section beam shown below has been designed to resist a torque, T, of 10 Nm. Evaluate the maximum shear stress, τ_{max}, and the relative rotation, θ, of the ends of the beam. (The shear modulus, G, of the beam material is 30 GPa.)



overall dimension not midline dimension

Let $\tau_{sz} = G\theta' \frac{dF}{dn}$ and $\tau_{nz} = -G\theta' \frac{dF}{ds}$

$\nabla^2 F = -2,\ \frac{dF}{ds} = 0$ and $F = 0$ at free surface

$\therefore \frac{\partial^2 F}{\partial n^2} = 0 \qquad \therefore F = -n^2 + \frac{t^2}{4}$

$\therefore J = \iint F\,ds\,dn = \iint_{-\frac{t}{2}}^{\frac{t}{2}} -n^2 + \frac{t^2}{4}\,dn\,ds = \frac{1}{3}\int t^3\,ds$

$= \frac{1}{3}\cdot(2mm)^3\cdot(2\cdot \overset{29}{30}mm + \overset{48}{50}mm) = \frac{880}{3}\,mm^4$

$\frac{848}{3}$

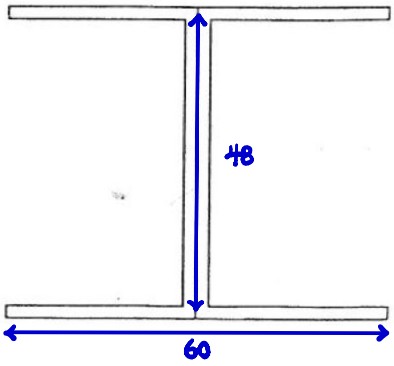
$\therefore \theta' = \frac{T}{GJ} = \frac{25}{22}\,m^{-1}$ $\quad \frac{125}{106}$

$\therefore \tau_{max} = |\tau_{sz}|_{n=\pm\frac{t}{2}} = |-2G\theta'(\pm\frac{t}{2})| = G\theta' t = 6.82\times10^7\,Pa$ $\quad 7.08\times10^7$

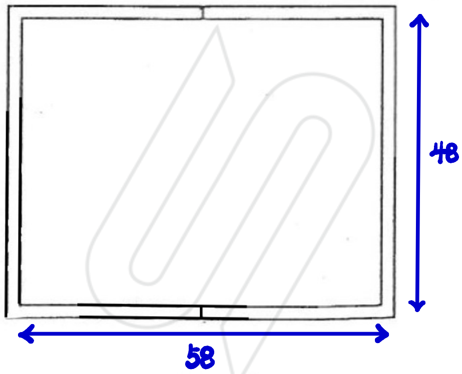
$\theta_{end} = \int_0^L \theta'\,dz = \theta' L = \frac{15}{22} = 0.682$ $\quad 0.708$

$\frac{75}{106}$

b) The beam is found to be too flexible in torsion and so the designer proposes two alternative schemes based on the original channel section. In scheme (i) two channels are adhesively bonded back-to-back. In scheme (ii) two channels are welded together along the edges of the flanges. Evaluate τ_{max}, and θ for each of these schemes. Discuss any assumptions you have made.



scheme i



scheme ii

For scheme i,

$$J_i = \frac{1}{3}\int t^3 ds = \frac{1}{3}\sum t^3 \cdot s = \frac{1}{3}\left[(2mm)^3 \cdot (4 \cdot 30mm) + (2 \cdot 2mm)^3 \cdot \overset{48}{50}mm\right] = \frac{4160}{3}\ mm^4 \quad 1344$$

$$\theta_i' = \frac{T}{GJ} = \frac{25}{104}\ m^{-1} \quad \frac{125}{504}$$

$$\therefore (\tau_{max})_i = \left|-2G\theta'(\pm\frac{t_{max}}{2})\right| = G\theta' t_{max} = 2.88 \times 10^7\ Pa \quad 2.98 \times 10^7$$

$$(\theta_{end})_i = \int_0^L \theta_i'\ dz = \theta_i \cdot L = \frac{15}{104} = 0.144 \quad 0.149 \quad \frac{25}{168}$$

For scheme ii, as closed-tube in torsion

$$\theta_{ii}' = \frac{1}{2AG}\oint \frac{q}{t} ds = \frac{T}{4A^2G}\oint \frac{1}{t} ds = \frac{T}{4(\underset{58 \cdot 48}{2 \cdot 30 \cdot 50})^2 G}\left[\frac{1}{2} \cdot (\overset{2 \cdot 58}{4 \cdot 30} + 2 \cdot \overset{48}{50})\right]$$

$$= 1.018 \times 10^{-3}\ m^{-1} \quad 1.140 \times 10^{-3}$$

$$\therefore (\tau_{max})_{ii} = \tau_{ii} = \frac{q}{t} = \frac{T}{2At} = 8.33 \times 10^5\ Pa \quad 8.98 \times 10^5$$

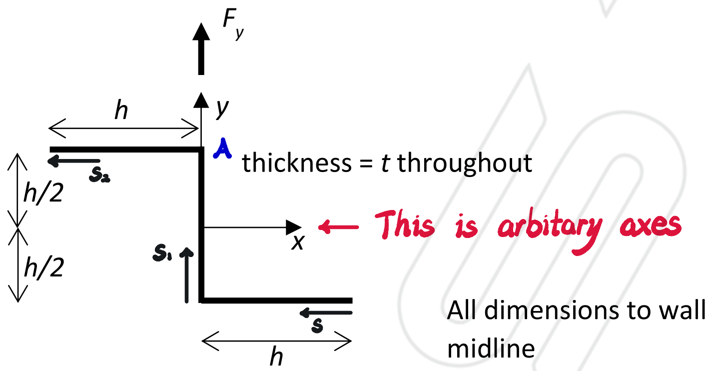
$$(\theta_{end})_{ii} = \int \theta_{ii}' \cdot dz = \theta_{ii}' \cdot L = 6.11 \times 10^{-4} \quad 6.84 \times 10^{-4}$$

Assumption

i: adhesive bond is sufficiently stiff to ensure the two webs of the channel section behave as a single web of thickness 4mm

ii: the welded connection is a full-thickness weld

2. Determine and sketch the shear flow in the lower horizontal wall of the thin-walled Z-section shown below due to the vertical shear force F_y .

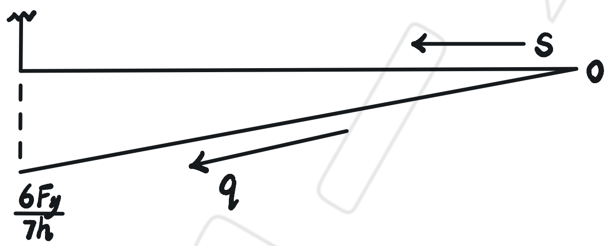


$$I_x = \int y^2\, dA = \frac{bh^3}{12} + \bar{y}^2 \cdot (bh)$$

$$= \frac{th^3}{12} + \left(\frac{h}{2} - 0\right)^2 \cdot th + \left(0 - \frac{h}{2}\right)^2 \cdot th = \frac{7}{12} th^3$$

$$(D_x)_s = -\int_0^s \bar{y}\, t\, ds = -\int_0^s \left(0 - \frac{h}{2}\right) t\, ds = \frac{1}{2} ths$$

$$\boxed{\therefore q_s = \frac{F_y}{I_x}(D_x)_s = \frac{12F_y}{7th^3} \cdot \left(\frac{1}{2} ths\right) = \frac{6F_y}{7h^2} s}$$



$$I_y = \int x^2\, dA = \cancel{\frac{t^3h}{12}}^{\,0} + \left[\frac{h^3t}{12} + \left(0 - \frac{h}{2}\right)^2 ht\right] + \left[\frac{h^3t}{12} + \left(\frac{h}{2} - 0\right)^2 ht\right] = \frac{2}{3} th^3$$

$$I_{xy} = \int xy\, dA = \left(0 - \frac{h}{2}\right)\left(\frac{h}{2} - 0\right) \cdot ht + \left(\frac{h}{2} - 0\right)\left(0 - \frac{h}{2}\right) \cdot ht = -\frac{1}{2} th^3$$

$$\therefore \bar{\bar{F}}_y = \frac{(F_y I_y - F_x I_{xy}) I_x}{I_x I_y - I_{xy}^2} = \frac{14}{5} F_y$$

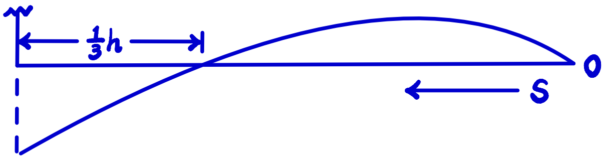
$$\bar{\bar{F}}_x = \frac{(F_x I_x - F_y I_{xy}) I_y}{I_x I_y - I_{xy}^2} = \frac{12}{5} F_y$$

$$(D_y)_s = -\int_0^s \bar{x}\, t\, ds = -\int_0^s (h - s)\, t\, ds = \frac{1}{2} ts^2 - ths$$

$$\therefore q_s = \frac{\bar{\bar{F}}_y}{I_x}(D_x)_s + \frac{\bar{\bar{F}}_x}{I_y}(D_y)_s$$

$$= \frac{24F_y}{5th^3}\left(\frac{1}{2}ths\right) + \frac{18F_y}{5th^3}\left(\frac{1}{2}ts^2 - ths\right)$$

$$= \left(\frac{9}{5h^3}s^2 - \frac{6}{5h^2}s\right)F_y$$



3. Verify that the x-coordinate of the shear centre of the above section is at x=0.

$$(D_x)_{s_1} = (D_x)_s\big|_{s=h} - \int_0^{s_1} \hat{y}t\,ds_1$$

$$= \frac{1}{2}th^2 - \int_0^{s_1} (s_1 - \frac{h}{2})t\,ds_1$$

$$= \frac{1}{2}th^2 + \frac{1}{2}ths_1 - \frac{1}{2}ts_1^2$$

$$(D_x)_{s_2} = (D_x)_{s_1}\big|_{s=h} - \int_0^{s_2} \hat{y}t\,ds_2$$

$$= \frac{1}{2}th^2 - \int_0^{s_2} (\frac{h}{2} - 0)t\,ds_2$$

$$= \frac{1}{2}th^2 - \frac{1}{2}ths_2$$

$$\therefore q_{s_1} = \frac{F_y}{I_x} D_{s_1} = \frac{12F_y}{7th^3} \cdot (\frac{1}{2}th^2 + \frac{1}{2}ths_1 - \frac{1}{2}ts_1^2) = -\frac{6F_y}{7h^3}s_1^2 + \frac{6F_y}{7h^2}s_1^2 - \frac{6F_y}{7h}$$

$$q_{s_2} = \frac{12F_y}{7th^3} \cdot (\frac{1}{2}th^2 - \frac{1}{2}ths_2) = \frac{6F_y}{7h} - \frac{6F_y}{7h^2}s_2$$

Take moment at origin.

$$F_y \cdot X_E = \int_0^h -\frac{h}{2} \cdot q_s\,ds + \int_0^h \frac{h}{2} \cdot q_{s_2}\,ds_2$$

$$= -\frac{3F_y h}{14} + \frac{h}{2}(\frac{6F_y}{7} - \frac{3F_y}{7})$$

$$= 0$$

$\therefore X_E = 0$ about origin

Take moment about A,

$$F_y \cdot X_E = \int_0^s -\frac{h}{2} \cdot q_s\,ds = -\frac{h}{2}\int_0^s (\frac{9}{5h^3}s^2 - \frac{6}{5h^2}s)F_y\,ds$$

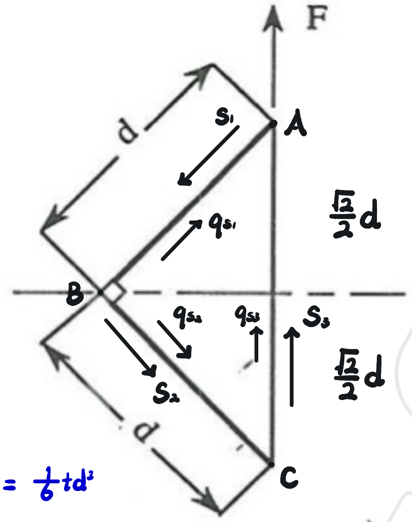
$$= -\frac{F_y}{2h^2}\left[\frac{3}{5}s^3 - \frac{3}{5}hs^2\right]_0^s$$

$$= 0$$

$\therefore X_E = 0$ about A

4. A thin-walled triangular section of constant thickness t is loaded by a shear force F along the hypotenuse as shown. Determine and sketch the shear flow distribution.

$I_x = I_{x_{AC}} + I_{x_{BC}} + I_{x_{AB}}$



where

$I_{x_{AC}} = \frac{t(\sqrt{2}d)^3}{12} = \frac{\sqrt{2}}{6} t d^3$ *

$I_{x_{BC}} = \int y^2 \, dA$ based on xy

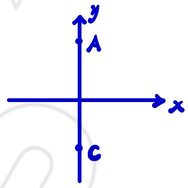
$= \int_{s_2} (s_2 \sin 45^\circ)^2 (t\, ds_2)$

$= \int_0^d \frac{1}{2} t s_2^2 \, ds = \frac{1}{6} t d^3$

$I_{x_{AB}} = \int_0^d (\frac{\sqrt{2}}{2} - s_1 \sin 45^\circ)^2 (t\, ds_1) = \frac{1}{6} t d^3$

$\therefore I_x = (\frac{1}{3} + \frac{\sqrt{2}}{6}) t d^3$

*



① $I_{x_{AC}} = \int y^2 \, dA = \int_{-\frac{\sqrt{2}}{2}d}^{\frac{\sqrt{2}}{2}d} \int_{-\frac{1}{2}t}^{\frac{1}{2}t} y^2 \, dx\, dy$

② $I_{x_{AC}} = \int y^2 \, dA = \int_0^{\sqrt{2}d} (s_3 - \frac{\sqrt{2}}{2}d)^2 (t\, ds)$

$$\boxed{\begin{aligned} I_x &= \int y^2\, dA = \frac{bh^3}{12} + \hat{y}^2 \cdot A \\ &= \frac{t(\sqrt{2}d)^3}{12} + 2t \cdot \int_{-\frac{\sqrt{2}}{4}d}^{\frac{\sqrt{2}}{4}d} y^2\, dy + (\frac{\sqrt{2}}{4}d - 0)^2 \cdot td + (0 - \frac{\sqrt{2}}{4}d)^2 \cdot td \\ &= \frac{\sqrt{2}}{6}td^3 + \frac{\sqrt{2}}{24}td^3 + \frac{1}{8}td^3 + \frac{1}{8}td^3 = (\frac{5}{24}\sqrt{2} + \frac{1}{4})td^3 \end{aligned}}$$

$(D_x)_{s_1} = -\int_0^{s_1} \hat{y} t\, ds_1 = -\int_0^{s_1} (\frac{\sqrt{2}}{2}d - \frac{\sqrt{2}}{2}s_1) t\, ds_1 = \frac{\sqrt{2}}{4}ts_1^2 - \frac{\sqrt{2}}{2}tds_1$

$(D_x)_{s_2} = (D_x)_{s_1}|_{s_1=d} - \int_0^{s_2} \hat{y} t\, ds_2 = -\frac{\sqrt{2}}{4}td^2 - \int_0^{s_2} (0 - \frac{\sqrt{2}}{2}s_2) t\, ds_2 = \frac{\sqrt{2}}{4}ts_2^2 - \frac{\sqrt{2}}{4}td^2$

$(D_x)_{s_3} = (D_x)_{s_2}|_{s_2=d} - \int_0^{s_3} \hat{y} t\, ds_3 = -\int_0^{s_3} (s_3 - \frac{\sqrt{2}}{2}d) t\, ds_3 = -\frac{1}{2}ts_3^2 + \frac{\sqrt{2}}{2}tds_3$

where $q = \frac{F_y}{I_x} D_x$

Take moment at A.

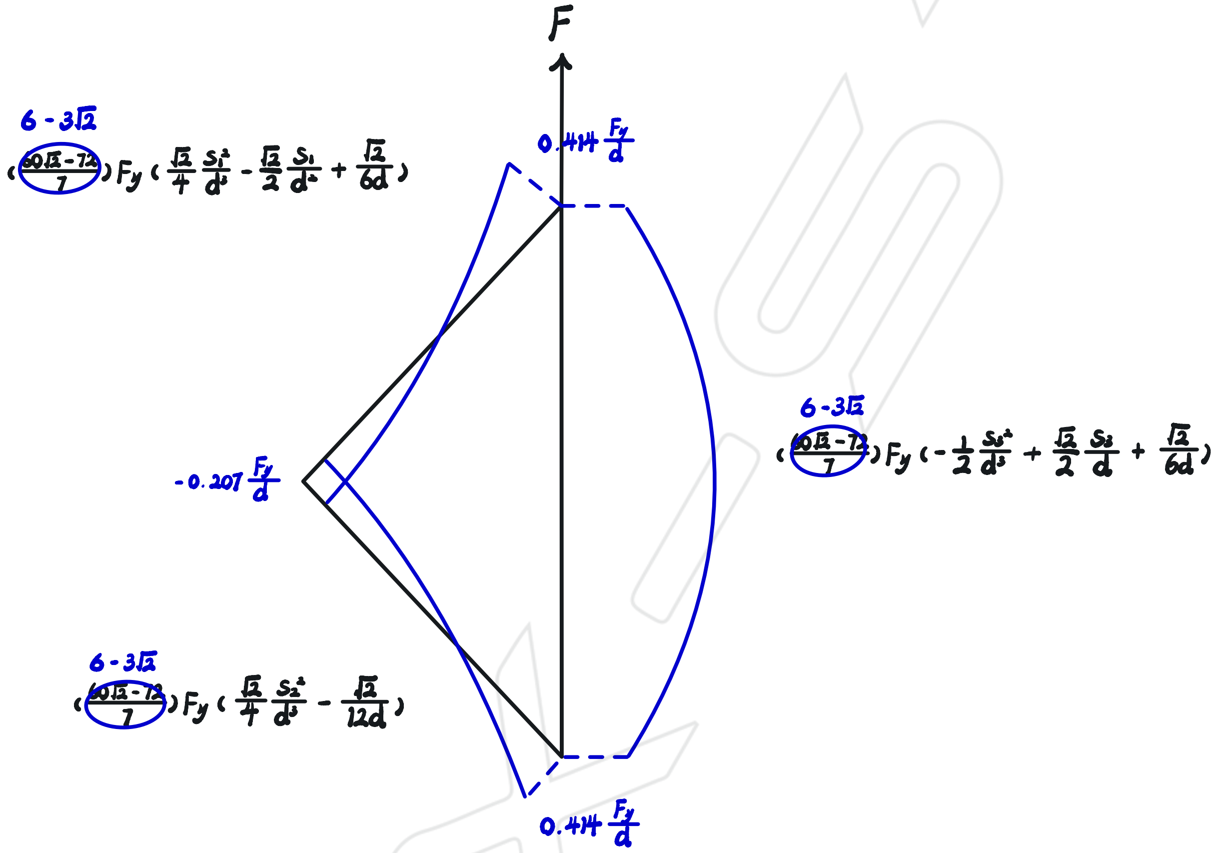
$\Sigma M_A(G) = d \cdot \int_0^{s_2} q_{s_2}\, ds_2 + 2Aq_0 = 0$

$\therefore 2Aq_0 = -\frac{F_y}{I_x} d \int_0^d \frac{\sqrt{2}}{4}ts_2^2 - \frac{\sqrt{2}}{4}td^2\, ds_2 = -\frac{F_y}{I_x} d (\frac{\sqrt{2}}{12}td^3 - \frac{\sqrt{2}}{4}td^3) = \frac{F_y}{I_x}(\frac{\sqrt{2}}{6}td^4)$

where $A = \frac{1}{2} \cdot d \cdot d = \frac{1}{2}d^2$

$\therefore q_0 = \frac{F_y}{I_x}(\frac{\sqrt{2}}{6}td^2)$

Therefore, $q = q_0 + q_b$



5. Idealise the section shown below as a boom-shear panel idealisation for bending about the principal y-direction. The booms are to be positioned at the ends of each straight wall segment.

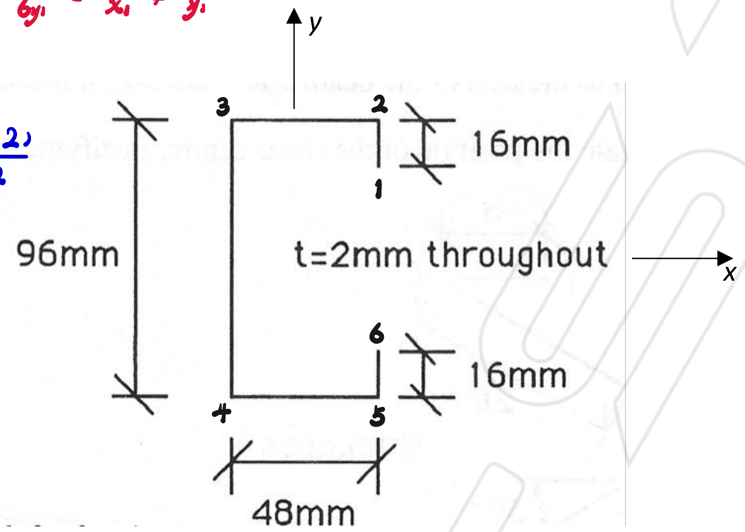
$\frac{\sigma_{y2}}{\sigma_{y1}} = \frac{x_2}{x_1} \neq \frac{y_2}{y_1}$

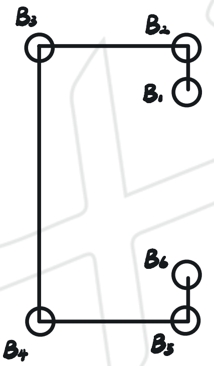
$\bar{x} = \frac{48 \cdot 2 \cdot (2 \cdot 16) + 24 \cdot 2 \cdot (48 \cdot 2)}{2 \cdot 2 \cdot 16 + 2 \cdot 48 \cdot 2 + 96 \cdot 2}$

$= \frac{7680}{448}$ (mm)

$= \frac{120}{7}$ (mm) $= 17.143$ (mm)

from 3-4





Due to symmetry, $B_1 = B_6$, $B_2 = B_5$, $B_3 = B_4$

$\Delta B_{1\to2} = \frac{tb}{6}\left(2 + \frac{y_2}{y_1}\right) = \frac{2 \cdot 16}{6}\left(2 + \frac{48}{32}\right) = \frac{56}{3}$ (mm²) $\qquad \frac{tb}{6}\left(2 + \frac{x_2}{x_1}\right) = \frac{2 \cdot 16}{6}\left(2 + \frac{\frac{216}{7}}{\frac{216}{7}}\right) = 16$ (mm²)

$\Delta B_{2\to1} = \frac{2 \cdot 16}{6}\left(2 + \frac{32}{48}\right) = \frac{128}{9}$ (mm²) $\qquad 16$ (mm²)

$\Delta B_{2\to3} = \frac{2 \cdot 48}{6}\left(2 + \frac{48}{48}\right) = 48$ (mm²) $\qquad \frac{2 \cdot 48}{6}\left(2 + \frac{-\frac{120}{7}}{\frac{216}{7}}\right) = \frac{208}{9}$ (mm²)

$\Delta B_{3\to2} = \Delta B_{2\to3} = 48$ (mm²) $\qquad \frac{2 \cdot 48}{6}\left(2 + \frac{\frac{216}{7}}{-\frac{120}{7}}\right) = \frac{16}{5}$ (mm²)

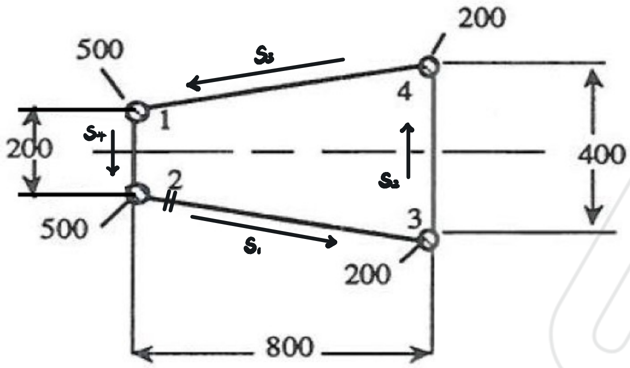
$\Delta B_{3\to4} = \frac{2 \cdot 96}{6}\left(2 + \frac{48}{-48}\right) = 32$ (mm²) $\qquad \frac{2 \cdot 96}{6}\left(2 + \frac{-\frac{120}{7}}{-\frac{120}{7}}\right) = 96$ (mm²)

$\therefore B_1 = B_6 = \Delta B_{1\to2} = \frac{56}{3}$ (mm²) 16

$B_2 = B_5 = \Delta B_{2\to1} + \Delta B_{2\to3} = \frac{560}{9}$ (mm²) $\frac{352}{9}$ $= 39.1$ (mm²)

$B_3 = B_4 = \Delta B_{3\to2} + \Delta B_{3\to4} = 80$ (mm²) $\frac{496}{5}$ $= 99.2$ (mm²)

6. The figure below shows a section which has been idealised as booms and shear panels. If all the shear panels have thickness t = 2 mm, locate the shear centre.



All dimensions in mm and to wall midline

$Y_E = 0$ at the horizontal line of symmetry and also $\bar{y} = 0$

$$\therefore I_x = (100-0)^2 \cdot 500 + (0-100)^2 \cdot 500 + (200-0)^2 \cdot 200 + (0-200)^2 \cdot 200$$

$$= 2.6 \times 10^7 \,(mm^4) = 2.6 \times 10^{-5} \,(m^4)$$

'Cut' the cross section at the point as shown above

$$\therefore (D_x)_{s_1} = -\int_0^{s_1} \bar{y} t \, ds_1 = 0$$

$$(D_x)_{s_2} = (0-200) \cdot 200 = 4 \times 10^4 \,(mm^3)$$

$$(D_x)_{s_3} = 4 \times 10^4 - (200-0) \cdot 200 = 0$$

$$(D_x)_{s_4} = 0 - (100-0) \cdot 500 = -5 \times 10^4 \,(mm^3)$$

$$\therefore q_{s_1} = \frac{F_y}{I_x} (D_x)_{s_1} = 0$$

$$q_{s_2} = \frac{F_y}{I_x} (D_x)_{s_2} = \frac{F_y}{650}$$

$$q_{s_3} = 0$$

$$q_{s_4} = \frac{F_y}{I_x} (D_x)_{s_4} = -\frac{F_y}{520}$$

At shear centre, $\frac{d\theta}{dz} = 0$

$$\therefore q_0 \oint \frac{1}{t} ds = \int_0^{s_1} \frac{q_{s_1}}{t} ds_1 + \int_0^{s_2} \frac{q_{s_2}}{t} ds_2 + \int_0^{s_3} \frac{q_{s_3}}{t} ds_3 + \int_0^{s_4} \frac{q_{s_4}}{t} ds_4$$

$$= 0 + \frac{4}{13} F_y + 0 - \frac{5}{26} F_y = -\frac{3}{26} F_y$$

where $\oint \frac{1}{t} ds = \frac{1}{2} (2 \cdot 806.23 + 400 + 200) = 1106.23$

$\therefore q_0 = -1.04 \times 10^{-4} F_y$

Take moment about 1.

$$F_y X_E = 2q_0 A + 800 \cdot \int_0^{s_2} q_{s_2} \cdot ds_2$$

$$= 2q_0 \left(\frac{200+400}{2} \cdot 800 \right) + 800 \cdot \frac{F_y}{650} \cdot 400$$

$$= 442.24 F_y$$

$\therefore X_E = 442.24$ (mm) to the right of 1-2